# For Developers

For the Spring 2026 semester, we will use a shared GitHub repository (folder). If you don't have a GitHub account, create one; either your personal account or your Illinois account should work. (I used my personal account because the data will persist after we graduate).

Our shared GitHub repository for the Quad Day project should be under the *ATLAS-Illinois VR* account. This repository is *forked* (derived) from Whirlwind31/ATLASQuadProject. (Whirlwind31 is my personal GitHub account).

When you're ready to develop for the project, first download GitHub Desktop. Then, **clone the Quad Day Project** using the **ATLAS Illinois** GitHub link above. Do *not* download as ZIP–then the local copy of your project will not link to GitHub.

Find the local folder for the Quad Day project in your File Explorer. Then, use Unity to "Add project from disk" and use the folder that is a direct parent of "Assets", "Scenes", "Scripts", etc. Finally, open the Quad Day project. Any changes that you make to the Quad Day project should now show up in the GitHub. For example, try renaming an object; that should show up as a change to the local scene.

Duplicate .gitignore upon cloning the new repository. Move this new .gitignore to the same repository as you use in the Unity editor. Do **not** commit the 32,000 temporary files that are generated upon starting the project for the first time.

Before you start developing, **copy the main scene** and rename the main scene with your name. For example, "QuadDay-AlexanderLau." We will all make changes to the same branch, but because everyone is working on different scenes, there should be few *merge conflicts.* (If we all edited the same scene and made separate changes from one another, it would be a hassle to decide which changes to keep & which to discard).

Continue developing on your personal scene from now on. You are now ready to develop content!

**Project Summary:**

Here are the current projects that can be expanded upon for future intern work:

1. Mammoth Tag / Mammoth Wandering AI

Synopsis: This game involves the player grabbing a golden donut from a mammoth's pedestal, followed by the mammoth chasing after the player. If the mammoth touches the player then the mammoth takes the golden donut back! But if you reset the game you can always try again.

<u>Idle State</u>

The mammoth stands on a wireframe pedestal and rotates to stare at the player. Technically the rotation script involves moving a very, very small distance towards the player, but in practice the mammoth doesn't move at this stage.

<u>How to Enrage the Mammoth</u>

There is a golden donut on a golden pedestal close to the mammoth with a sign that says (paraphrased) "Mammoth's property: Do not touch." If you approach the pedestal (i.e. walk close to it), then the mammoth goes into an "enraged state." This is visually indicated when the sign changes text to "RUN!!!!!"

More specifically, if you reach within 2 distance units from the pedestal then the enrage script triggers. It is *not* dependent on the time you first grab the golden donut. Feel free to change this; there have been several times where I have been unable to grab the golden donut before the mammoth reaches it first.

<u>Enraged Mammoth</u>

The mammoth will begin to run towards the player, like a game of tag. More specifically, the mammoth chooses a point in the direction of the player and runs towards that point before braking, coming to a full stop at that point, and then choosing another point. Oftentimes that point can overshoot the player, which is fine because it means the mammoth will be at top speed when it hits the player and triggers a game over.

Your objective is to keep the donut away from the mammoth's collider (and yourself! If the mammoth hits you then it's game over as well).

The mammoth has two settings. The current implementation is called "Aries," where the mammoth tends to change course quickly. Generally this can be implemented by decreasing the "AvoidDistance," which means that the mammoth will move forward less before changing its direction to face the player again. This means you can outrun the mammoth at times just by running straight.

“Taurus” involves the mammoth moving very quickly towards a point that is far away from the mammoth (and oftentimes the player). It is prone to overshooting the player and so it is easy to avoid by sidestepping the mammoth. The player must still be careful because the point could be slightly behind the player, and the player must then sidestep again once the mammoth reaches said point. However, this is generally an easier AI than “Aries.”

The mammoth has a pathfinding algorithm that allows it to avoid obstacles. It is not perfect and near the corners of the map this can lead to some odd behavior. In general, directing the mammoth to the map’s corners makes its pathfinding less effective, although it’s also more unpredictable.

<u>Game Over State</u>
The mammoth takes the donut onto its tusk and reverts to its idle state of staring at the player and rotating to face the player’s direction. The golden donut on the mammoth’s tusk is purely cosmetic and is a *different donut object* from the one that the player holds. The tusk donut does not have collision.

Currently this game is **incomplete.** There are a few known bugs and glitches related to the mammoth tag project, which include:

1. <u>The mammoth can push the player through several building walls.</u> This can trap the player inside said building. For example, I managed to replicate this behavior with the English Building and Illini Union; the latter building is unused and so the floors currently flicker. This can be changed by moving the Illini Union’s y position up by 0.01, but the real issue is the softlock. The player is then unable to get out.

2. <u>When the mammoth collides with the player, the C# script sometimes does not detect a losing condition.</u> Normally this should trigger a Game Over, where

   Not doing so means that the mammoth could push the player for long periods of time before the losing script finally triggers and the Game Over sequence occurs. This could exacerbate issues like (1).

3. <u>There is leftover code from an earlier iteration of the mammoth script in NavMeshMovement.cs.</u> This iteration dates back to around c. Summer 2024. You are free to experiment with that code and see what the earlier script is like. Alternatively, you can clone the repository at the end of Fall 2025 to see the mammoth’s original behavior!

4. <u>The game over animation is incomplete.</u> I was hoping to implement a “scripted cutscene” where the mammoth returns the golden donut to the pedestal and then the mammoth returns to its wireframe starting position. However, I ran out of time before I could do so. The main objective of this game over animation would be to allow the player to try again, because the current implementation only gives the player one chance at the mammoth game before having to reset the Unity project.

This is a more advanced task because it involves coordinating several elements of the project together and a lot of programming in C#.

2. Archery Game

Worked on by Victor Liu (Spring 2026), Sathvik Lanka (Fall 2025)

3. Parkour Game

Worked on by Carlton Woo (Spring 2026)

4. Meta XR Assistant

Worked on by Mike Lee (Spring 2026)

5. POINT VR - Sandbox

Worked on by Alexander Lau (Summer 2025, Spring 2026)

6. Scavenger Hunt

Worked on by Alexander Lau (Summer 2025, Fall 2025, Spring 2026) - Needs regression testing

7. Alma Mater Ring Toss

Not worked on since Spring 2025, although there was a documented bug throughout Fall 2025 where the rings would disappear from the Unity project

8. **Fixing Mismapped Building Textures (more research-based)**

Worked on by Alexander Lau (Fall 2025)

9. Illini Logo Arrangement

Worked on by Carlton Woo (Fall 2025)

**Spring 2026 (updated May 15, 2026):**

Here is a list of bugs, glitches, & unintended interactions between projects that could motivate VR work next semester.

Please also refer to the project documentation for Victor Liu (archery game), Mike Lee (XR assistant), Carlton Woo (parkour game), if available.

1. <u>Archery Game + Mammoth: Arrows remain stationary while mammoth moves</u>

Shooting arrows at the mammoth reveals that the arrows stay in place at their target upon collision. When the mammoth moves around from its original spot, the arrows are left floating in midair.

2. <u>Archery Game: Quiver does not properly snap to user's hip</u>

This may have been an issue while merging all other VR interns' contributions together. The quiver does not seem to be able to attach to the user's hip; upon releasing the grip, the quiver drops to the floor rather than to the hip as expected. There may be a different attachment point than the hip, but I am not certain.

3. <u>Archery Game: Quiver can be "double grabbed" and moved around</u>

If you grab the quiver with one controller and attempt to grab it again with another, then you can freeze the quiver in place occasionally. Moving around and then releasing one of the hands results in you being able to move around the quiver from a faraway distance (as if you were grabbing it from close by), resulting in a "telekinesis" effect.

4. <u>Archery Game: Quiver collides with other objects (e.g. U of I logos)</u>

Not necessarily a bug, but in general we must be wary about whether or not certain objects should be able to have collisions with one another. Combined with (3), you can use the quiver to knock into blocks in the "Block I" game. This can lead to some weird physics interactions, notably when pushing the blocks into a wall or into the floor (the blocks may even deform as a result of this behavior).

5. <u>XR Assistant: Disclaimer, need to connect API key for demos</u>

For the Assistant and Captions translation to work, you need to connect an API key, likely from Google Gemini, into the Unity scene. This is particularly important if we want to demo the full Quad Games.

6. <u>XR Assistant: Camera app's photos are significantly darker</u>

It seems that the camera is unable to capture the full lighting of the Quad Games scene, resulting in photos in the Gallery appearing significantly darker than they do in-game.

7. <u>XR Assistant: Camera interface does not move with user's vision</u>

This means that you cannot take a picture of the sky, because the camera is locked to pointing slightly towards the ground. If we could configure this so that the camera interface could face any direction, we could then take pictures as necessary.

8. XR Assistant: UI Menu is very close to user

It is close enough so that the user can reach out and touch the menu to navigate, although this also restricts movement for the user if they do not wish to interact with the AI assistant features in the Quad Games.

9. Parkour Game & Scavenger Hunt: Control conflicts

Parkour game: A to jump, X to vault (vaulting only possible on ladder)
Scavenger hunt: A / B to turn pages of paper (right hand), X / Y on left hand

You can stay on the ladder and vault while turning the page (which is likely less of an issue).

The real issue is jumping every time you wish to turn the page forwards on the paper, which hurts the scavenger hunt. The fix would involve disabling jumping while the scavenger hunt paper is being held (most likely by using a public variable that checks whether the player is grabbing any piece of paper, and then disabling jumping when that occurs. As the player can hold two papers at once this is an intermediate-level task).

# Scavenger Hunt Walkthrough

For ATLAS Quad Games

This is for the purposes of documentation. If you don't want to be spoiled about the contents of the Scavenger Hunt, don't read beyond this page!

Verbal explanation (as of December 17, 2025)

1. **Beginning**


   a. There is a tree in the part of the Quad opposite the Illini Union, where Foellinger Auditorium is supposed to be. The big Lincoln Bust is nearby.
   b. ‘The tree has a signpost saying “Come Inside!” from an inconvenient direction. Go inside the tree.

2. **Piece of Paper**

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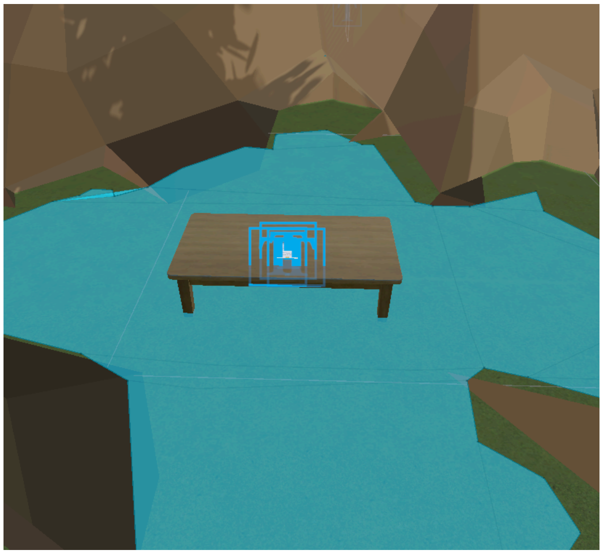
|---|---|
|  | • You should see a table with a piece of paper. Pick up the piece of paper with the left grip or right grip button. This paper is special:<br><br>• If you pick up the paper with the left grip button, use the X / Y buttons to turn the page forward / backward.<br>• If you use the right grip button, use the A / B buttons to turn the page.<br>• You can only turn the page if you're currently holding it.<br>• If you tilt the page left and right, the paper's text will disappear. This is an intentionally coded "invisible ink" effect. A previous intern commented that it seemed like a bug, but fixing this bug would be very complicated. |

3. **Note: Hints**
    a. Hints are provided for the player at the back of the paper. There are multiple hints to each section. I designed the hints this way on purpose so experienced players can complete the scavenger hunt without any hints, while beginner players can read the hints if they get stuck.

4. **Solution to Part 1**

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|  | There's a tree in front of the Illini Union. *This is not the correct tree.* |

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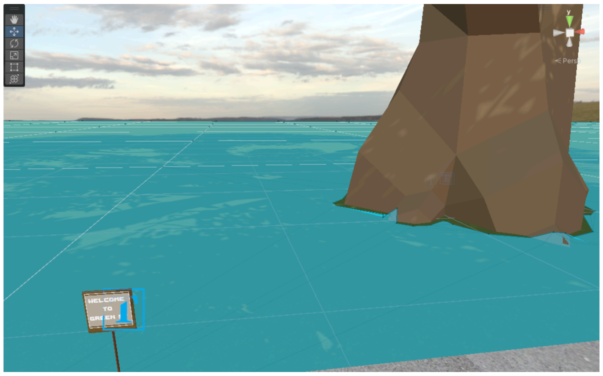
|  | While holding the paper containing Part 1's instructions, navigate behind the Illini Union. If you see a signpost saying "Welcome to Green St.", you are on the right track.<br><br>The next part of the scavenger hunt is hidden in the giant tree behind the "Welcome to Green St." signpost. |

5. **Coin Puzzle**

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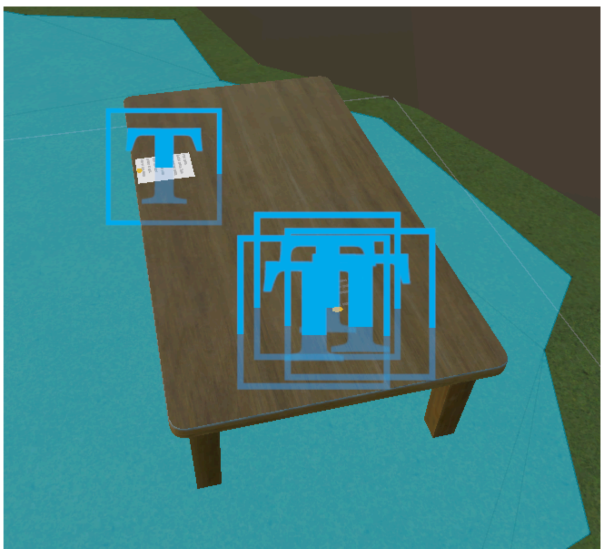
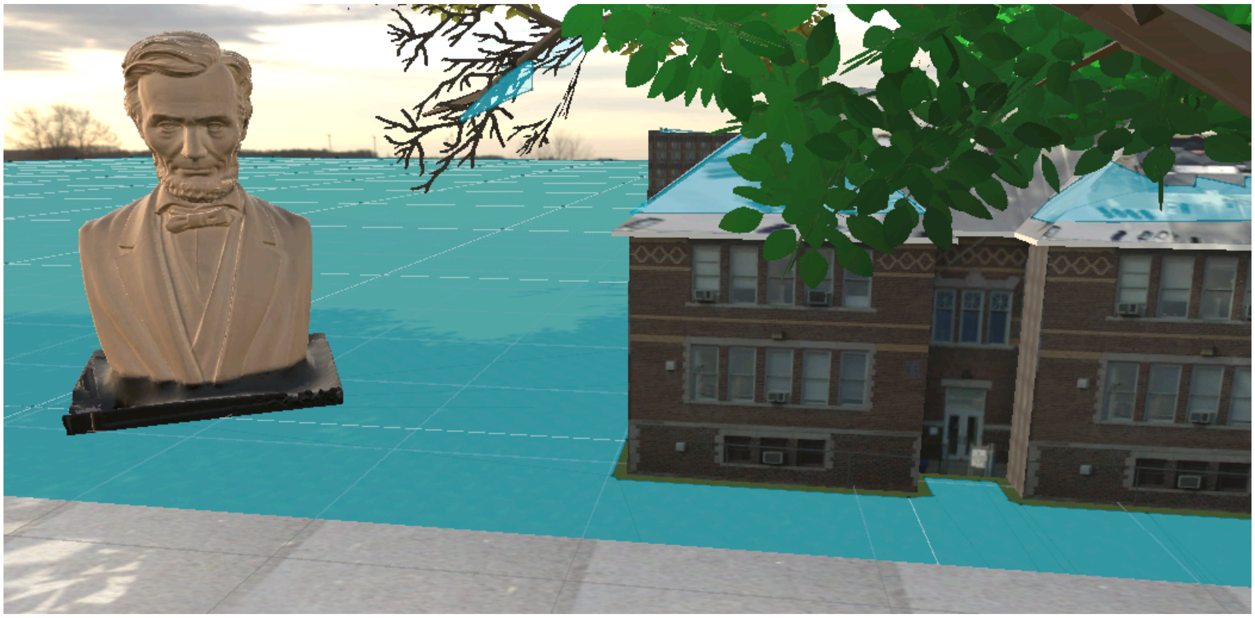
|  | You should see a poem about coins. There is also a coin in the back right of the table. Flip over the coin to reveal the instructions to Part 2. |

6. **Solution to Part 2**



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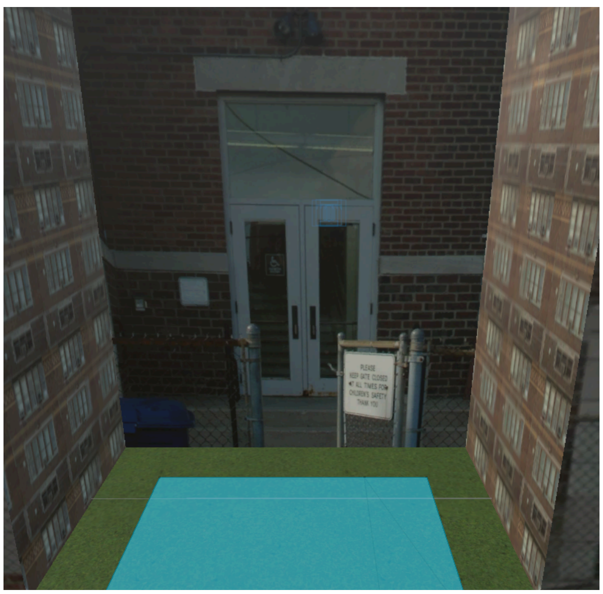
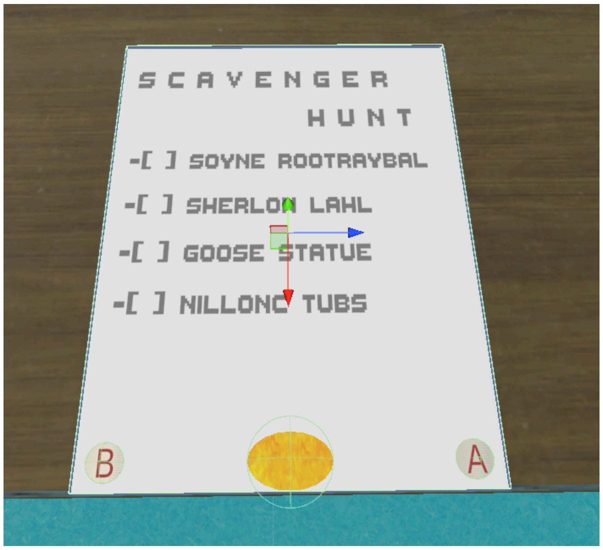
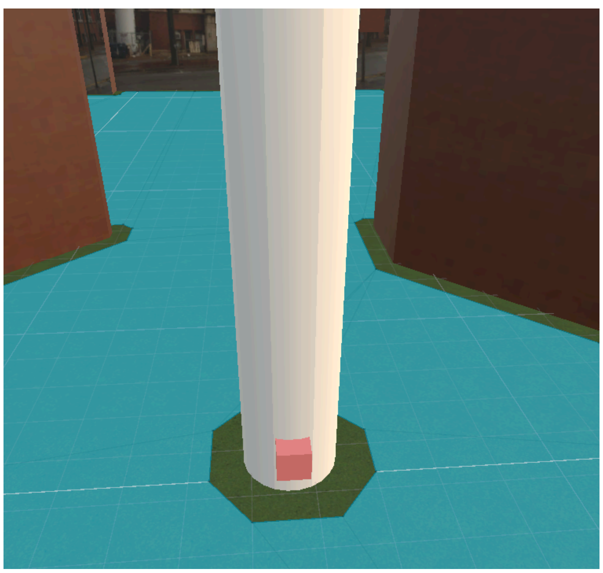
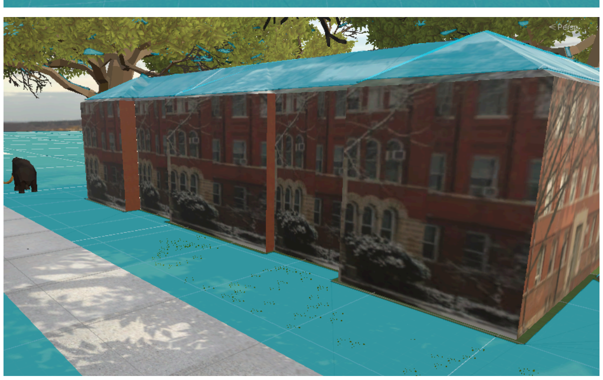
|  | Bring Part 2’s instructions to the front of the Colonel Wolfe School, which is the building right next to the giant Lincoln Bust. You should see that the front of the gate clearly says “keep gate closed for children’s safety.”<br><br>Walk into the building, where you should see a table with the instructions to Part 3 already face-up. |
|  | There is a building right next to the Colonel Wolfe School with three columned pillars. This is the English Building and is not the correct answer.<br><br>Reason for misconception: Some playtesters thought the pillars looked like a gate. |

### 7. Solution to Part 3




Don’t lose Part 3’s instructions! There should be a page with a checklist on it. You’ll have to turn to the page of the checklist every time you want to check something off.





First enter the building closest to the mammoth’s spawn location. This is **Noyes Laboratory**. Travel in front of Noyes Laboratory (specifically, in the center facing the Quad). Then walk straight until you see a red box attached to a column.

If you turn to the page of the checklist, you will unlock the next location and its respective red box. The red box should turn green if done correctly, and the checklist will display the unscrambled name with a ‘X’ in the checkbox.

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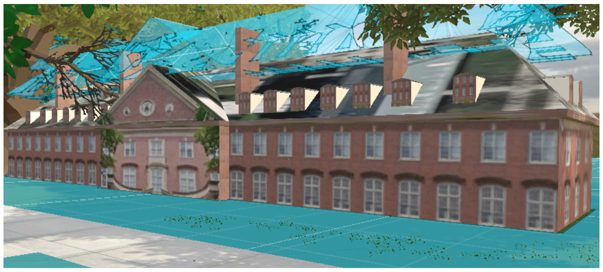
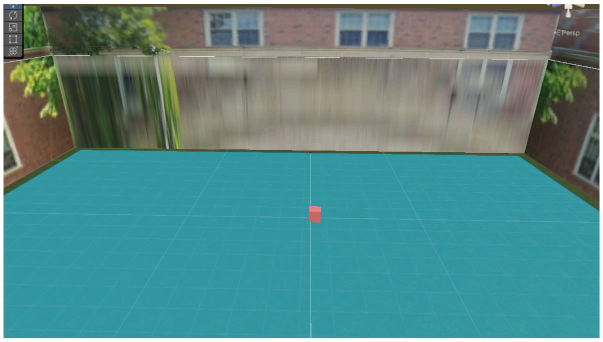
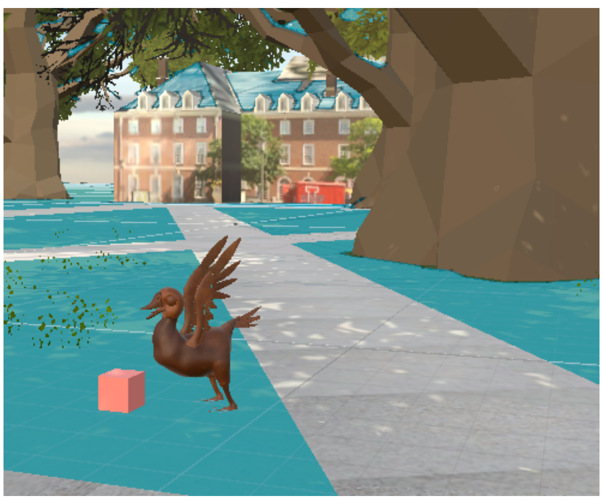
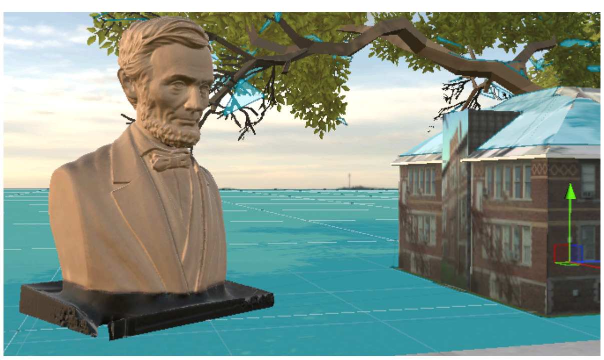
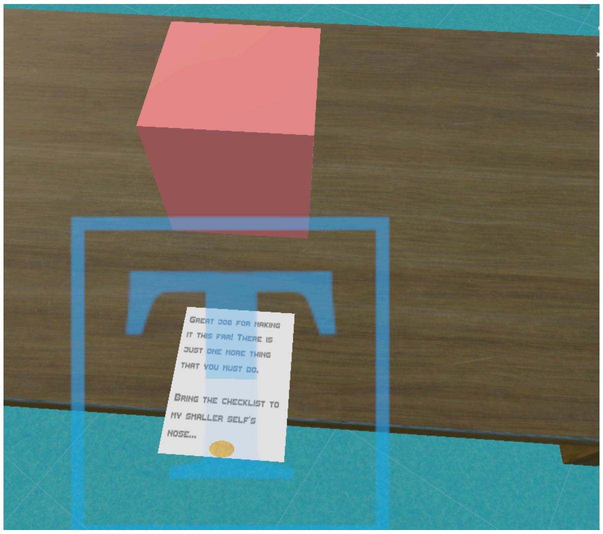
|   | Next, visit **Wohlers Hall.** This is the building right next to Noyes in VR Quad Day. There are two rooms in Wohlers Hall; enter through the front, then pass through the *middle of the wall* in front of you. |
|  | There is then a **Goose Statue** outside, in the Main Quad. Horizontally it is between Noyes and the Main Quad, but vertically it is close to the center of the Quad. Approach the red box. |
|  | Finally, visit the big **Lincoln Bust** once more next to the Colonel Wolfe School. If done correctly, you should see a one page document informing you that you must *bring the checklist to "his smaller self's nose."* |

<table>
<tr>
<td>
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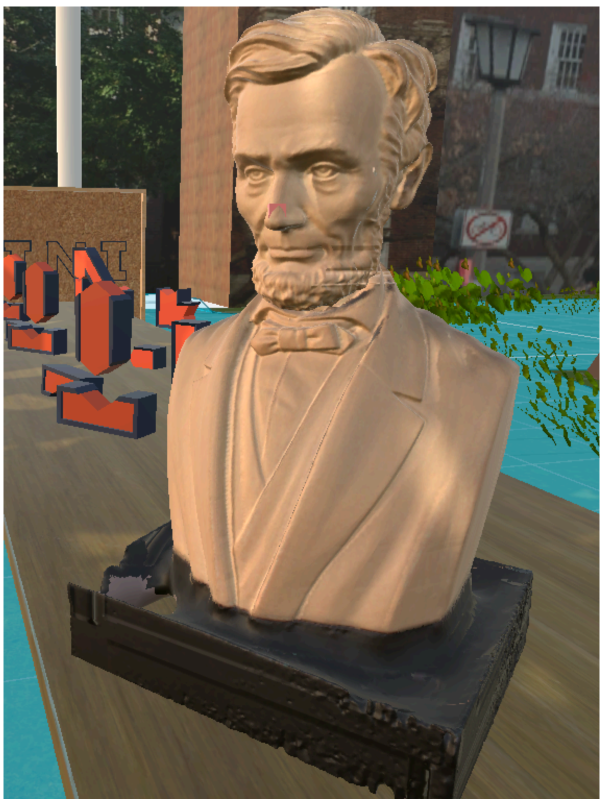
<tr>
<td></td>
<td>In front of the English Building, close to your spawn location, there's a small Lincoln Bust. Approach the small Lincoln Bust with your checklist from Part 3. It should have a red box in its nose that turns green.<br><br>With that, you have completed the Scavenger Hunt! The final secret is inside the Main Library.</td>
</tr>
</table>

**8. Secrets of the Main Library**

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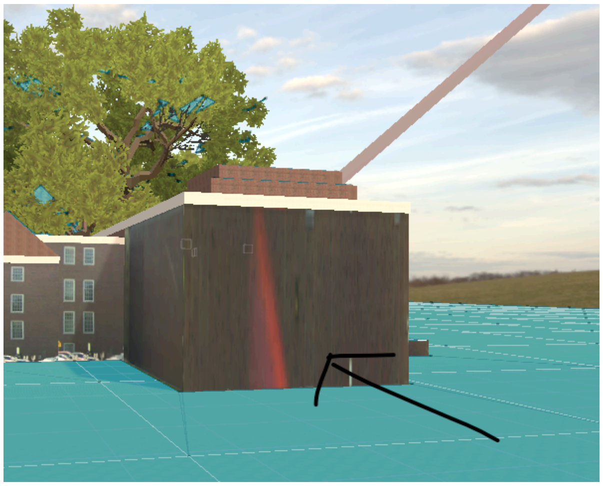
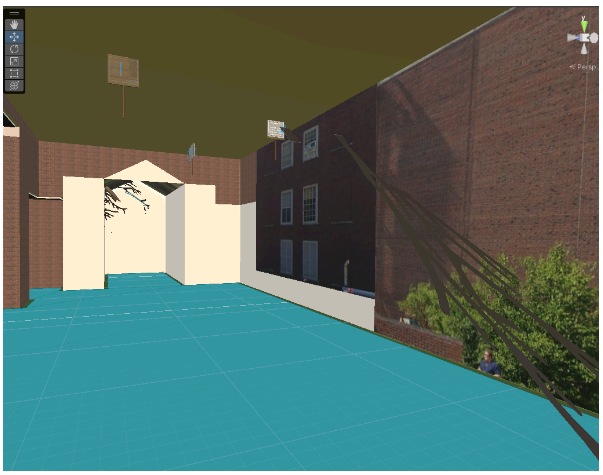
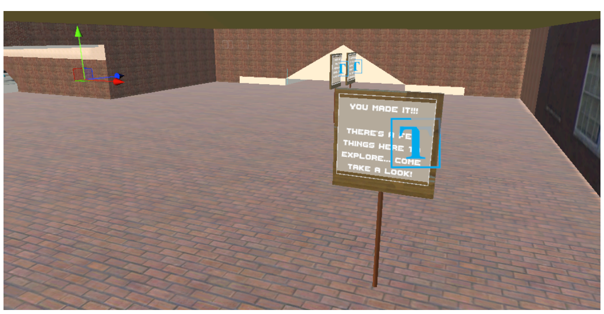
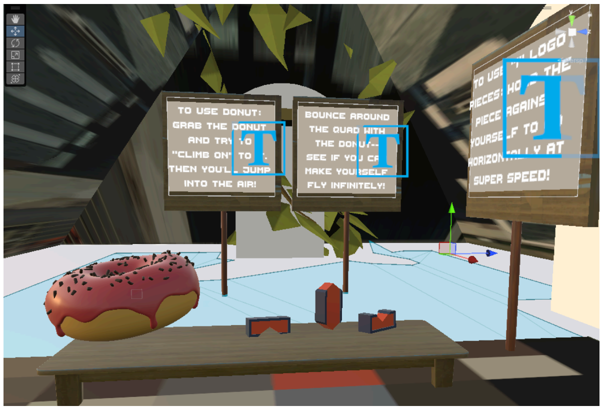
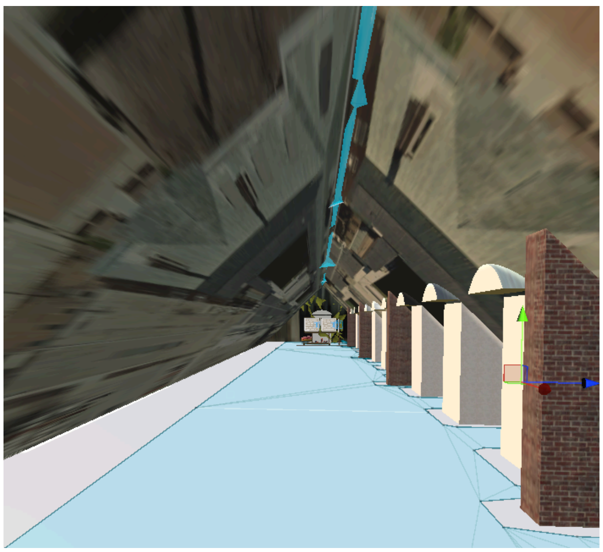
|  | Most players will have to explore around the Main Library a little bit to find this secret.<br><br>Go to the right wall of the Main Library. You'll see a wall with a bright red light shining down on it. Enter that way. |
|  | You should see a large hallway with floating signposts. These signposts will indicate to general players that this is the room with the secret.<br><br>There's a twig that leads upward. Climb the twig to uncover the second floor of the Main Library! |
|  | Here you can explore the "attic" of the library. Some parts look kind of like a cathedral. |
|  | Watch out for trees; specifically, this one in the middle of the second floor. If you get stuck in a tree you may have to redo the entire scavenger hunt!<br><br>(There are outside sections to the library floor as well. Check them out, but avoid the trees). |





Inside one of the Main Library's "attics" you will find a donut and Illini logo floating in space. With the donut, you'll be able to fly if you move forward while your legs are touching the donut.

I found the donut to allow you to bounce very high into the air, sometimes infinitely if you're lucky. The logos, on the other hand, allow you to move extremely quickly in a horizontal direction.

a. You *can* bounce with the Illini logo, and move horizontally with the donut. But I found vice versa to be far more effective personally.

Again, take care not to get stuck in trees, especially if you're using the donut. If you do you'll have to restart the scavenger hunt.

(Also, I found the donut escaping during my playtests and bouncing down to the ground. Unfortunately I don't have a fix yet).

### 9. Secret Observatory


You may have also noticed another tree trunk leading upward. Climb it to reveal a chamber with one final table and piece of paper. **Don't climb if you have a fear of heights!**

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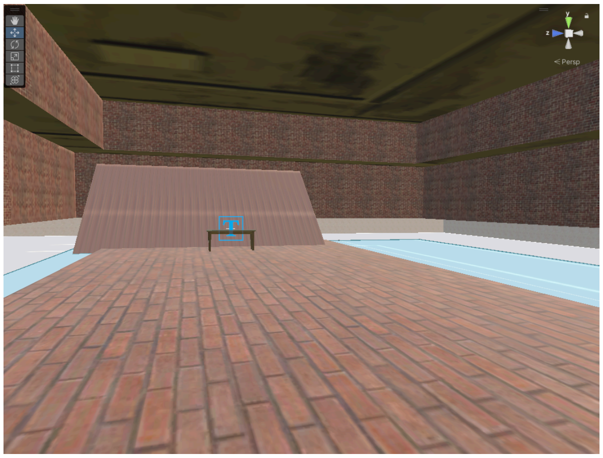
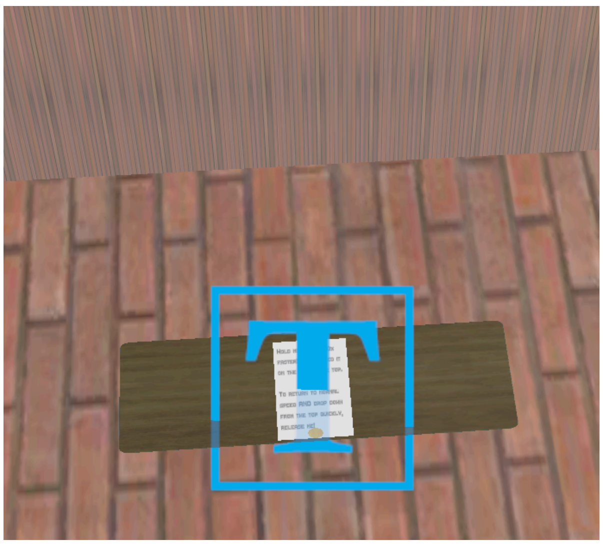
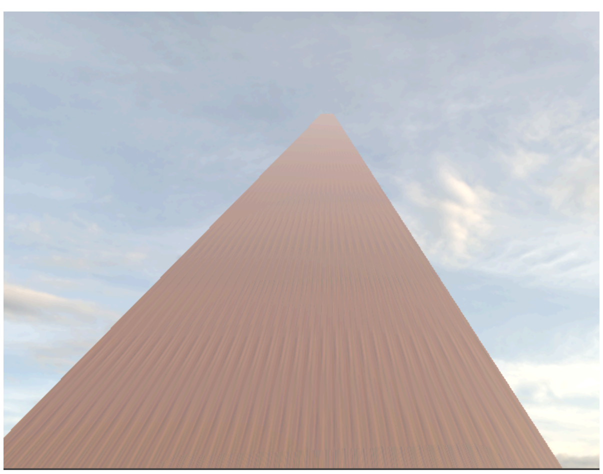
| <br> | The piece of paper serves two purposes. One, you will move 40 times faster. Your acceleration will increase from 0.25 to 10, allowing you to move lightning fast. (It's hard to control though - use at your own risk!)<br><br>The other purpose is to stand on top of the observatory. If you let go of the paper, you will fall through the floor of the observatory. This is intentional, so that you can explore the other elements of Quad Day whenever you choose. |
|  | Climb the sloped pathway up to the observatory. There should be guard rails that protect you from falling. (Don't go too fast or else you might run through the guard rails. This can either lead you to fall back to the ground, or into the abyss if you ran far enough! Watch out!) |

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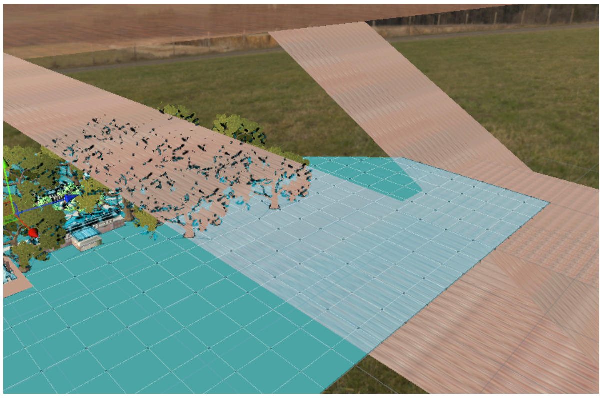
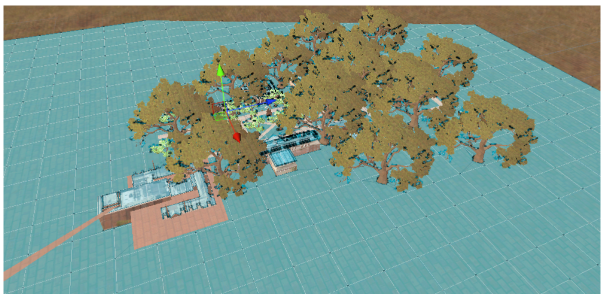
|  | Fair warning: There’s a flat section. Once you reach there, stop and turn left once you see a diagonal turn. Then climb the remaining slope to reach the observatory! |
|  | Eventually you’ll reach the observatory. If you brought the piece of paper, you can stand on top of the observatory; otherwise you cannot. Now enjoy the sights that you managed to uncover!<br><br>(Left is a view from the observatory. The observatory is just a floor with 5% opacity. Isn’t that cool?) |